Partial Derivatives and the Gradient

First and second partial derivatives

1 For $f(x, y) = x^3y^2 - 4xy + y$, find:

a f_x

b f_y

c f_{xx}

d f_{xy}

2 For $f(x, y) = e^{xy}\sin(x + y)$, find f_x using the product rule.

The gradient and directional derivatives

3 Find the gradient ∇f of $f(x, y) = x^2y + 3y^3$ at the point $(1, 2)$.

4 Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4\rangle$.

Tangent planes

5 Find the equation of the tangent plane to $z = f(x, y) = x^2 + y^2$ at the point $(1, 2, 5)$, using $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

6 Find the equation of the tangent plane to $z = xy$ at the point $(2, 3, 6)$.

Critical points of multivariable functions

7 Find the critical point(s) of $f(x, y) = x^2 + y^2 - 4x + 2y + 5$ by setting $f_x = 0$ and $f_y = 0$.

8 Verify that the critical point found above is a minimum, using the second derivative test with $D = f_{xx}f_{yy} - (f_{xy})^2$.

The chain rule for multivariable functions

9 Let $z = x^2y$, where $x = t^2$ and $y = t^3$. Use the multivariable chain rule $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ to find $\frac{dz}{dt}$ at $t = 1$.

10 Let $w = x^2 + y^2 + z^2$, where $x = \cos t$, $y = \sin t$, $z = t$. Find $\frac{dw}{dt}$ using the chain rule.

Maximum rate of increase

11 Find the direction in which $f(x, y) = x^2 + xy$ increases most rapidly at the point $(1, 2)$, and state the maximum rate of increase.

12 Find the direction of steepest descent for $f(x, y) = x^2 + y^2$ at the point $(3, 4)$.

Implicit partial differentiation

13 For the surface $x^2 + y^2 + z^2 = 9$, find $\frac{\partial z}{\partial x}$ by implicit differentiation (treating y as constant).

Classifying critical points with saddle points

14 Find and classify the critical point(s) of $f(x, y) = x^2 - y^2$, using the second derivative test.

15 Find and classify the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.

An applied optimization problem

16 A company's profit is modeled by $P(x, y) = 100x + 120y - x^2 - 2y^2 - xy$, where x and y are units of two products. Find the production levels that maximize profit by setting $P_x = 0$ and $P_y = 0$.

Higher-order mixed partial derivatives

17 For $f(x, y) = x^3y^2 + \sin(xy)$, find f_{yx} and f_{xy}, and verify they are equal (as guaranteed by Clairaut's theorem for continuous mixed partials).

Linear approximation in two variables

18 Use the linear approximation $f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$ to estimate $f(2.1, 2.9)$ for $f(x, y) = x^2y$, using the point $(2, 3)$.

19 The volume of a cylinder is $V(r, h) = \pi r^2 h$. A cylindrical can is manufactured with radius $r = 3$ cm and height $h = 10$ cm, but due to manufacturing tolerances, the radius might be off by as much as 0.05 cm and the height by as much as 0.1 cm. Use the total differential $dV = V_r\,dr + V_h\,dh$ to estimate the maximum possible error in the computed volume.

20 For $f(x, y) = x^2y - 3xy^2 + y^3$, find each of the following at the point $(1, -1)$:

a $f_x(1, -1)$

b $f_y(1, -1)$

c The gradient vector $\nabla f(1, -1)$

d The directional derivative in the direction of $\langle 1, 0 \rangle$